SUNIL KUMAR

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SUMMARY

CSAI undergraduate at IIIT Delhi experienced on building real-world ML and backend systems. Good at developing scalable applications,consistently optimizing for real-world latency and accuracy tradeoffs, not just benchmark scores.

EDUCATION

B.Tech in Computer Science and Artificial Intelligence (CSAI) 2022 - 2026
Indraprastha Institute of Information Technology (IIIT Delhi), New Delhi

CBSE Class 12 2022
New Era Public School, Dwarka Percentage: 96

CBSE Class 10 2020
New Era Public School, Dwarka Percentage: 88

SKILLS

ML/Data: PyTorch, TensorFlow, Scikit-learn, XGBoost, OpenCV ,SHAP
Languages: Python, Java, C++ , Swift
Backend: Django, FastAPI, Node.js, AWS
Databases: MongoDB, PostgreSQL
Frontend: React.js, Next.js, Tailwind CSS
Tools: Git, Docker, Linux, PDB , Windsurf , PowerBI , Postman

EXPERIENCE

Lab@IIIT-Delhi, UG Researcher July 2025 - May 2026
Advisor: Prof. Pankaj Vajpayee Track: Entrepreneurship

- Designed a full-stack trust-based marketplace with idempotent APIs, role-based auth and multi-channel (SMS/WhatsAp job-lifecycle notifications. Engineered a 6-factor function $S(t,r) = w_d f_d + w_s f_s + w_b f_b + w_p f_p + w_r f_r - w_l f_l$ and a Bayesian-smoothed reputation system with recency decay to address cold-start and rating-gaming risks
- Led primary research: 580-respondent survey across rural/semi-urban India, uncovering that 88% of repair requests flow through informal channels vs. 12% to authorised service centers.
- Validated end-to-end workflows via a simulated pilot; defined a KPI framework (target: $<$60min response time, $>$85% completion rate, $\geq$4.2/5 technician rating) as the success bar for future field deployment

PROJECTS

Meesho Multi-Label Fashion Attribute Classification - Github Jan 2025 - Apr 2025
Python, PyTorch, EfficientNet-B5, Django, React, Typescript

- Built a 5-head EfficientNet-B5 classifier predicting 10 attributes across 5 clothing categories on 70K images; achieved 73% harmonic F1-score with $<$1s inference latency via GPU-accelerated fine-tuning

WiFi Indoor Localization & Navigation System - Github Nov 2025 - Mar 2026
Python, Flask, A Pathfinding, KNN, Sensor Fusion (PDR + Kalman)*

- Mapped a 125m $\times$ 22m floor as a 68-node, 70-edge graph; A* pathfinding verified across 35 destinations with 100% path accuracy . Fingerprinted 35 grid points across 15+ APs; fused KNN (K=3) with PDR + Kalman over 2–4s scan intervals via a 6-endpoint Flask API

Diabetes Risk Prediction Using BRFSS - Github Nov 2025 - Dec 2025
Python, Scikit-learn, XGBoost, CUDA, SHAP

- End-to-end XGBoost pipeline on 70K samples, 23 features; got 0.8246 ROC-AUC on Tesla T4; reduced feature space 34.8% (23→15) with only 1–2% accuracy drop via Random Projection

Recom-System using Neural Matrix Factorization - Github Jan 2026 - Mar 2026

Python, PyTorch, NumPy, Pandas, Scikit-learn, Pearl, CSS

- Benchmarked 6 RecSys models on MovieLens 100K; NeuMF with NDCG@10: 0.4503, outperforming popularity baseline by 53.6%
- Deployed hybrid GMF+MLP Flask API serving 943 users across 1,682 movies with 4 top-k inference strategies

CERTIFICATIONS

Self-Driving Cars Specialization View

University of Toronto (Coursera)

Built perception, state estimation, and motion planning pipelines; implemented vehicle localization using an Error-State Extended Kalman Filter, lane detection and trajectory generation with velocity .

Generative AI in Action View

IBM SkillsBuild

Designed prompt pipelines and LLM-integrated workflows for text classification, summarization, and retrieval-augmented generation across real-world use cases.

GPU Programming View

High Performance Computing

Implemented parallel CUDA kernels; optimized thread-block configurations and memory access patterns to minimize latency on large-scale workloads.

AI for Scientific Research View

Automated research workflows

ADDITIONAL INFORMATION

Languages: English (Professional), Hindi (Native)

Interests: Autonomous systems, generative AI applications, game development, automotive design